

where you will find a baseball player, then you go right, where you will find a harvestman, then you go right, where you will find a neck brace. Now you have all the information on the map. You start at the position where the torch is located, then you go right by one step, then you go right by one step, then you go up by one step, then you go up by one step, then you go left by one step, then you go down by one step, and then you go down by one step. What will you find?

Response Example See Figure 15.

C Visual State Tracking

As for where this emergent ability stems from, it might derive from tabular data, city grid navigation, maze exploration related coding problems [YBL⁺23]. These tasks involves understanding and manipulating objects in a 2D square grid. Besides, we conjecture the exposure of ascii-art comments [Reg19] during LLMs’ code pre-training possibly enhances this generalized ability. As a fact to support this conjecture, the visual tiling task is different from navigation tasks because it requires shape understanding and spatial manipulation ability. While tabular data and square grid navigation data boost row-wise or column-wise attention, ascii-art supplements intricate spatial attention to understand and manipulate 2D shapes. Additionally, ascii-art in code comments is presented in various formats, one of which is interleaved ascii diagrams, natural language and programming language. It require LLMs to generate the interleaved *mental images* and text sequence, thereby enhancing spatial visualization ability and spatiotemporal causality. Interestingly in the natural language navigation task, when GPT-4 is prompted with "use ascii-art to visualize", the complete tracking rate increases to 98.5% (+78.5%), boosting task performance to 62.5% (+3.5%).

C.1 Ascii-art in Code Comments

Ascii-art is commonly used in code comments to represent data structure, diagram, geometry and so on, which could benefit LLMs’ spatial understanding and visualization capability. Besides, it’s also used to illustrate how an algorithm works or simulate an operation, where reasoning traces and corresponding visualization are presented in an interleaved manner. Below are several examples in open-source projects.

- **Spatial Causality:** [Double-ended queue in Rust](#), [Scrolling web pages](#) and [tree rotation](#) present triplets of previous visual state, instruction, and updated state of instruction following.
- **Temporal Causality:** [Undo systems from emacs](#) provides various temporal states of the undo system when undo operation happens in different timelines and corresponding visualizations in an interleaved manner. Each visualization reflects the temporal causality of the system state.

This kind of interleaved sequence tracks the system state over time, thus reflecting spatiotemporal causality.

D Performance Trends Across Levels

In this analysis, we examine performance trends across varying difficulty levels in the next-step prediction task for models utilizing either CoT or VoT methods. These trends are crucial for understanding the inherent unpredictability associated with random guessing. As k increases from 2 to 7 in a k -step navigation map, distinct performance patterns emerge among different models, as depicted in Figure 9. Larger language models such as GPT-4 and LLAMA3-70B demonstrate a more predictable decrease in accuracy with increasing k . This trend indicates a robust ability to handle progressively challenging tasks, despite the overall decrease in performance. Detailed statistics are provided in table 6. In contrast, less powerful models like GPT-3.5 and LLAMA3-8B exhibit an irregular performance trajectory. These models show variable accuracy, with significant fluctuations at higher difficulty levels, suggesting a reliance on random guessing, particularly under conditions of increased task difficulty. This behavior highlights their limitations in sustaining reliable reasoning processes through more complex scenarios. Furthermore, the VoT method seems to offer a modest improvement in performance for the less powerful models, particularly in scenarios of lower difficulty.

This observation suggests that VoT might be advantageous for enhancing reliable reasoning in simpler spatial reasoning tasks, potentially compensating for the inherent weaknesses of smaller language models.

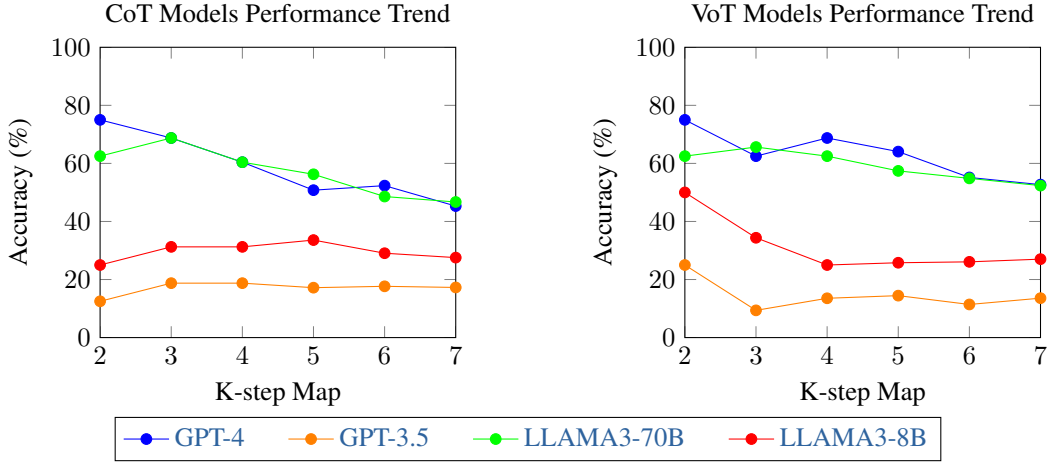


Figure 9: Performance Trends of CoT and VoT Models Across difficulty levels in next-step-prediction task.

Model	K-step Map	Map Count	CoT Accuracy (%)	VoT Accuracy (%)
GPT-4	2	8	75.00	75.00
	3	32	68.75	62.50
	4	96	60.42	68.75
	5	256	50.78	64.06
	6	640	52.34	55.16
	7	1488	45.30	52.69
LLama3-70B	2	8	62.50	62.50
	3	32	68.75	65.63
	4	96	60.42	62.50
	5	256	56.25	57.42
	6	640	48.59	54.84
	7	1488	46.71	52.35

Table 6: CoT and VoT performance of advanced models in next-step-prediction task across various difficulty levels. While performance drops as difficulty level increases, VoT method generally maintains a higher accuracy compared to CoT, highlighting its robustness in more challenging scenarios.

E Case study

We consider visual state tracking similar to spatiotemporal simulation. During the simulation in those tasks, we discovered several interesting behaviors of LLM.

1. Diverse visualization formats for state tracking: Nearly 30 different symbols found in the navigation tasks to track the navigation progress, including marking the path, marking the current location. Among those diverse representations, LLM succeeded in some challenging cases where it used directional arrow emojis to indicate both the location and moving direction at each step. More examples could be found in Appendix E.1.

2. Inconsistency between language and visualization: This is commonly observed across all tasks. Due to the limited visualization capability, sometimes LLM generates accurate language instruction but inaccurate visualization. And in other cases, LLM generates wrong answers even the visualization is generated correctly, which reflects its limitation of spatial understanding as discussed in previous section. More examples could be found in Appendix E.2.

3. Self-refine mechanism: We found several cases in visual tiling tasks where spatial hallucination happens due to the inconsistency or inaccurate visualization. Subsequently, LLM refined its reasoning, resulting in an accurate visualization and the correction of the final answer. More examples could be found in Appendix E.3.

E.1 *mental images* for State Tracking

In the visual navigation task, LLM adopted various symbols and representations to track the state of navigation progress. As shown in Figure 10, there're several tracking styles.

- Mark the path: adopting an identical symbol to mark current location or part of the path.
- Mark path and direction: using directional arrows to mark current location and indicate the moving direction simultaneously, which is more challenging than simply marking the path.
- Mark path with temporal steps: using numbers to demonstrate both temporal steps and current location.
- Remove road: turning roads into obstacles to avoid turning back, instead of adopting additional symbols to mark the path.

E.2 Inconsistency between Language and Visualization

In the visual tiling task, two inconsistent steps are highlighted in Figure 16. One is the inconsistent visualization with the language instruction of "place Variation 6 of Tetromino L". Another is the wrong decision to chose "Variation 2 of Tetromino I" given the visualization of the valid state.

E.3 Self-refine Mechanism

We found visualization could enhance LLM's reasoning by self-grounding and refining subsequent reasoning steps in some cases. As shown in Figure 17, despite successfully identifying variation 1 of tetromino L as incorrect option, GPT-4 excluded the correct option of variation 6 even it's placed accurately due to spatial hallucination (overlapping with yellow pieces), which led to a impossible solution. Then it detected the mistake and re-evaluate the placement of variation 6. Finally it placed the correct piece into the top left corner and validated the answer by filling the remaining space.